Executable Operating Contracts for Evidence Reuse in Aircraft Localization and Inspection

Chuansu Peng[a], Haoran Dong[a], Ruoxuan Wu[a], Peipei Zeng[b]

[a]*School of Aeronautical Engineering, Civil Aviation University of China, China*
[b]*Engineering Training Center, Civil Aviation University of China, China*

Abstract

Engineering changes alter the statistical evidence applicable to an inspection output. We introduce executable operating contracts that connect changing aircraft-inspection requirements to task-specific numerical services and reusable risk statements. A typed metamodel and unit-aware request compiler bind events, policies, source versions, producers and calibrated cards. Event containment governs proof reuse, requested-function invariance governs producer transfer, and persistent source history governs issuance. A reuse-first route with exact-event fallback preserves all 59 directly supported request contexts under fixed numerical policies. Its selected proof catalogue requires 63,488 event–record evaluations rather than 73,728, a 13.89% reduction. Paired geometry, raster and decoder re-execution on five archived inputs checks producer changes independently of declared dependency rules and flags two deliberately omitted coupling classes. A separate deterministic matrix verifies 15,360 protocol decisions. The underlying evaluation includes 13,824 independent synthetic localization episodes, 36 complete-route inputs and two image-task studies with 12,800 independent inputs. Calibration-scope sensitivity tightens example risk margins while leaving primary-allowance request support unchanged. The resulting interface coordinates evidence reuse, direct renewal, selective withdrawal and restoration across localization, acquisition, association and reporting.

Keywords: Executable operating contracts, Evidence reuse, Anytime localization, Risk calibration, Aircraft inspection

1. Introduction

Engineering changes alter the relationship between an inspection output and the evidence supporting its use. A consumer may relax an image requirement, a decoder may be replaced, or a newer localization estimate may become available after capture. Each change poses a concrete question: which existing statistical statements remain applicable, which must be withdrawn, and what computation is required to restore the requested service? Aircraft inspection provides a connected setting in which geometric estimation, image acquisition, association and reporting depend on different parts of the same information pipeline.

Anytime computation connects estimation quality to processing time [1, 2]; engineering knowledge models connect decisions to requirements and dependencies [3, 4, 5]. Risk-controlled selection attaches statistical evidence to a specified effective program [6]. An inspection service must preserve all three relationships through execution and change. Successive registration checkpoints revise one immutable scan/prior, capture consumes one installed version, and a risk card concerns a policy, population, processing service and failure event.

We introduce executable operating contracts for evidence reuse in aircraft localization and inspection (Fig. 1). A typed task–source–evidence representation compiles the consumer's requirements into executable predicates and a committed computation plan. Event-containment witnesses identify covered requirements; requested dependency projections identify applicable producer snapshots. Ordered source replacement and a persistent execution-history monitor connect these decisions to the output actually produced. The resulting interface supports evidence reuse, selective withdrawal and restoration across changing inspection requirements.

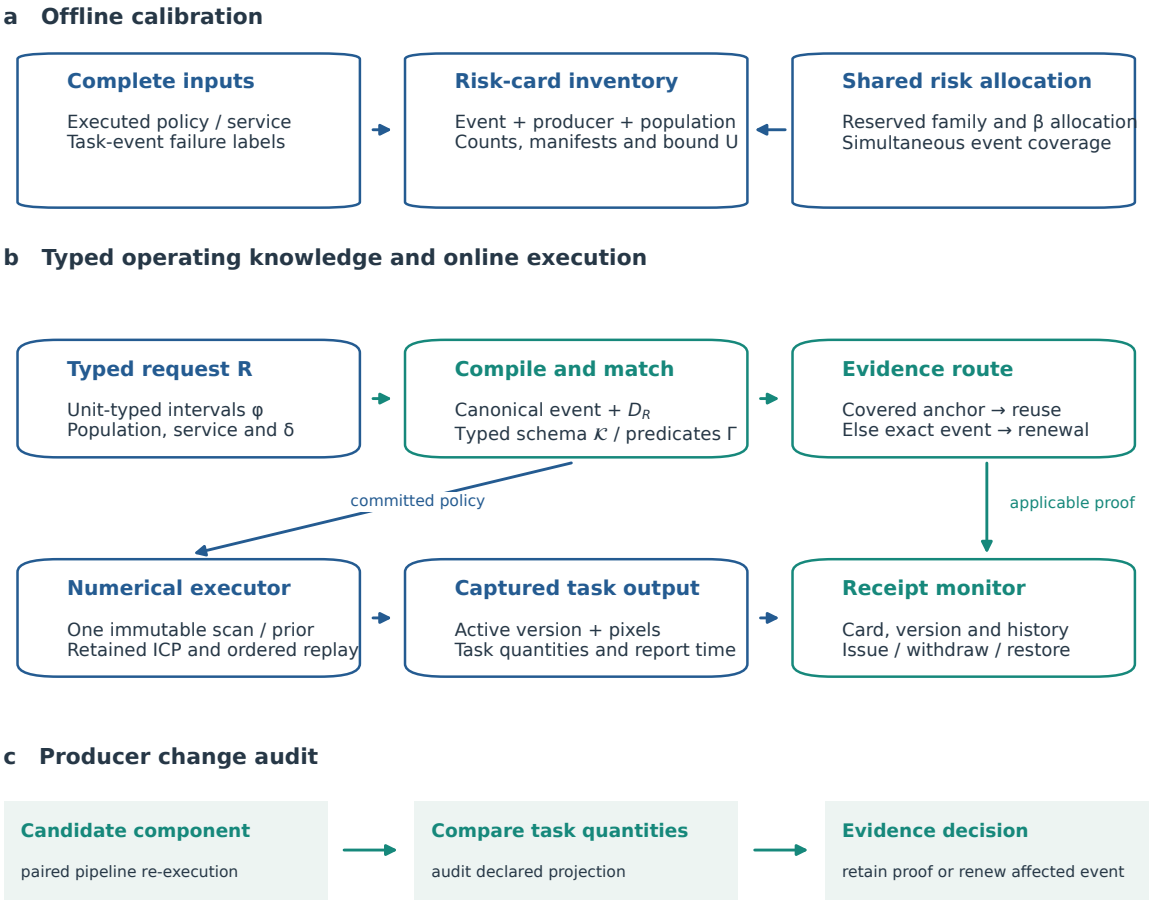


Figure 1: Architecture of the executable evidence interface. (a) Calibration records supply event-specific cards; typed requirements compile into a committed policy and evidence plan. (b) Source-consistent numerical execution supplies capture-time quantities and ordered versions. (c) Producer auditing, dependency projections and source-history checks govern reuse, direct fallback and issuance. The metamodel makes these relations explicit; its predicates perform the decisions.

The paper makes three contributions. First, it defines a typed metamodel and request compiler connecting task events, effective policies, numerical versions, producer dependencies and calibrated cards. Second, it establishes executable admission, transfer and issuance rules, including a reuse-first evidence route with direct renewal as its fallback. Third, it evaluates numerical services, producer-function changes and evidence construction through complementary designs. Paired pipeline re-execution tests the consequences of declared and deliberately omitted dependencies. On the same 59 supported contexts and fixed numerical policies, the reuse-first route retains direct-renewal support while reducing selected proof-catalogue evaluations from 73,728 to 63,488. Deterministic protocol checks separately verify state and attachment implementation.

The article develops the representation and execution rules, then evaluates request support, preparation work and change decisions alongside the underlying numerical services.

2. Related work

2.1. Engineering knowledge and change operations

Process knowledge and decision knowledge graphs organize engineering choices and their supporting context [3, 4]. Semantic change propagation makes component relations available for impact analysis [5, 7], while inspection ontologies connect operating conditions to robot requirements [8]. Their contribution is assessed through engineering operations: representational coverage and queries [9], or executable conversion and downstream model fidelity [10]. Our corresponding operation is the continued applicability of a statistical statement after a task or provider changes. It requires both the quantity's implemented dependency projection and the failure event certified by the original card. A component-impact relation becomes actionable when it identifies the certified event, applicable population and permissible evidence transfer. These links define the engineering operation studied here.

2.2. Computation contracts and source revisions

Metareasoning and anytime algorithms allocate computation according to its expected value [11, 12, 13]. Pant et al. connect computational contracts to estimation and control [1]; Aldana-López et al. study stability-preserving perception scheduling [2]. We retain ICP and joint-state estimation [14, 15, 16], using complete-event calibration alongside approximate curvature covariance [17, 18, 19]. The interface tracks correlated checkpoints as revisions of one source, then links the version used at capture to the requested evidence. This source identity is carried through replacement and causal motion replay.

2.3. Risk selection and evidence lifetime

Assumption–guarantee contracts formalize component obligations [20]; binomial inversion and risk-controlled selection calibrate specified configurations [21, 6, 22]. We use simultaneous finite-family coverage to support before-sensing policy and anchor selection, then extend each anchor through explicit containment and requested-function invariance. Runtime assurance supplies assumption monitoring and controller switching [23]. Here the monitored action is evidence issuance, with a source-scoped history latch that persists through plan reconstruction while numerical installation continues through its original interface.

Table 1 identifies the links introduced by the operating contract relative to these established operations. The central contribution is an executable evidence lifecycle: task/event identity, producer applicability and numerical source history jointly govern admission, retention, withdrawal and restoration. The evaluation measures those operations directly through matched preparation counts and change decisions.

Table 1: Positioning by engineering operation. Each row maps an established research operation to the executable relation evaluated here; it is a conceptual comparison of scope.

Established operation	Executable relation introduced here	Evaluation endpoint
Knowledge and change reasoning [3, 4, 5]	Requested quantities link provider changes to specific certified events	Affected statements withdrawn; applicable statements retained and restored
Anytime estimation and scheduling [1, 2]	Immutable-source revisions link a committed policy to the version consumed at capture	Source trace, localization trade-offs and complete-route availability
Risk-controlled configuration selection [6, 22]	Containment and invariant requested projections link new requests to existing risk cards	Unique request support and preparation work on matched contexts
Runtime assumption monitoring [23]	Persistent source history links attachment attempts to evidence status	Invalid-packet refusal, gate reconstruction and trusted-scope recovery

Table 2: Typed operating records and their executable roles. Events, quantities and provider identifiers supply the nested metamodel.

Record	Required specification	Executable decision
Request/event	Population, service, unit-typed intervals, allowance	Compile and check the required event
Policy/service	Effective program, action rule, budget, load, deadline	Commit a policy before sensing
Source/version	Immutable scan/prior key, sequence, payload, seal	Replace the prior contribution and replay motion
Producer/quantity	Quantity functions and declared provider snapshot	Audit changes and compare requested projections
Evidence card	Event, program, population, source manifests, n, k, U	Validate the bound and containment witness
Plan/receipt	Request, policy, card, source version, persistent history	Issue, withdraw or restore applicable evidence

3. Operating contracts for anytime localization

3.1. Typed task–source–evidence representation

The operating knowledge structure is

$$\mathcal{K} = (\mathcal{V}, \tau, \mathcal{R}, \sigma, \Gamma), \tag{1}$$

where entities $\mathcal{V}$ have types τ, relations $\mathcal{R}$, immutable specifications σ and executable predicates Γ. The core types are request, event, policy, source version, producer, quantity, provider, card, plan and receipt. Relations have explicit domains and ranges: a request *requires* an event, a card *bounds* that event and *supports* a plan, a plan *commits* a request and *selects* a policy, and a receipt *uses* a source version. Quantities *depend on* providers. A *revises* edge requires the same immutable source key and a strictly increasing sequence. The supplement gives the complete relation schema and executable instance checks.

A request $R = (g, a, \varphi, \delta)$ fixes population, service, a predicate conjunction and risk allowance before sensing. Its grammar is

$$\varphi ::= \mathrm{Interval}(m, u, \ell, h) \mid \varphi \wedge \varphi, \tag{2}$$

where m identifies a registered quantity, u its engineering unit and omitted bounds represent unbounded sides. The compiler checks dimensions, converts bounds to base units, intersects repeated intervals, rejects contradictory specifications and sorts quantities into a canonical event identity. It returns the required event and the union of its provider dependencies. Nonfinite required quantities fail the compiled event. The executable compiler reproduces all 24 declared task/variant events, including their dependency projections, and checks invalid units, bounds and relation types.

The reference implementation declares quantity dependencies in a versioned manifest derived from implemented data flow. Compilation aggregates those declarations; producer re-execution audits their numerical consequences. The representation supplies a typed execution and evidence interface, with predicates for unit validity, policy/population matching, containment, dependency projection, active source version and persistent history. The admission rule is

$$\begin{aligned} &\mathrm{ValidCard}(A) \wedge \mathrm{Match}(A, R, \pi) \wedge \mathrm{Cover}(A, R) \\ &\quad \wedge \mathrm{SameProjection}_{D_R}(p, p_A) \wedge U_A \le \delta \implies \mathrm{Admissible}(\pi, A, R). \end{aligned} \tag{3}$$

The following sections define containment, transfer and source-history semantics. Runtime issuance additionally requires a valid source seal, the active ordered version and a valid persistent history. These predicates determine inspectable admission, withdrawal and restoration actions.

3.2. Immutable sources and retained computation

The robot localizes against a stationary airframe prior. The joint state $x = (t, R, b, m_t, m_R)$ contains robot translation and rotation, six motion-bias components and map translation and rotation, with an 18-dimensional local covariance P. Its airframe-relative position error is

$$e_p = \widehat{m}_R^\top(\widehat{t} - \widehat{m}_t) - (m_R^\star)^\top(t^\star - m_t^\star). \tag{4}$$

Measured motion propagates the joint state and covariance using an $SE(3)$ linearization [16]. Simulator truth supplies offline labels; policy decisions use causal estimates and declared specifications.

A source s consists of a scan and its immutable prior (x_s^-, P_s^-). Robust point-to-plane registration minimizes

$$\Phi_s(x) = d^\top (P_s^-)^{-1} d + \sigma^{-2}\left[\sum_{i\in\mathcal{I}(x)} \rho(r_i(x)) + (n - |\mathcal{I}(x)|)\rho(r_{\text{out}})\right], \quad d = x \boxminus x_s^-. \tag{5}$$

The unmatched-point penalty keeps difficult correspondences in the objective. Each rolling scan is motion compensated and voxel filtered, retaining up to 1,500 frontend points. The primary solver then uses at most 96 deterministically thinned points from this local visible patch, 25 iterations, Huber scale 0.0625 m, $\sigma = 0.025$ m and $r_{\text{out}} = 0.8$ m. Checkpoints occur at iterations 1, 2, 4, 8, 16 and 25 or termination. An eligible candidate has a descending step, at least 12 pairs, clipped residual RMS below 0.065 m and more than 60% of residuals within 0.10 m. Final acceptance also requires translation and rotation steps below 0.001 m and 0.0005 rad. Supplementary methods specify covariance, backtracking and policy details.

Continuation retains the iterate, prior precision and computational prefix. At delivery time t_d, a candidate replaces the active version of its source and replays only acquired motion:

$$(x_{t_d}^{(j)}, P_{t_d}^{(j)}) = \mathcal{R}_{s\to t_d}(x_s^{(j)}, P_s^{(j)}; u_{s+1:t_d}). \tag{6}$$

Checkpoint revisions share one likelihood and prior. The receiver consequently replaces them in order; final rejection can restore the original source prior through the same interface. The serial service permits one active source job and drains pending publications before releasing it. Under a common retained solve and no interleaved independent update, delivery of the final version produces the same state as delivering that version alone at the same time. Figure 2 connects an archived numerical trace to the downstream task and its separately checked evidence.

3.3. Events, policies and computation budgets

For a four-second episode, integrated position loss is $L_\pi = \int_0^4 \|e_p^\pi(t)\|^2 dt$, evaluated by a 0.1-s right-endpoint sum. At study threshold $\epsilon = 0.10$ m, define

$$E_{\text{tr}} = \{\max_{t\in\{0.5,0.6,\ldots,4.0\}} \|e_p(t)\| > \epsilon\}, \tag{7}$$

$$E_{\text{obs}} = \{\text{no positive source version at } D\} \cup \{\|e_p(D)\| > \epsilon\}, \quad E_{\text{un}} = E_{\text{tr}} \cup E_{\text{obs}}. \tag{8}$$

Same-tick ordered installation and rollback precede assessment. Trajectory, observation, both-marginal and union requests use these distinct events. A union request uses its directly evaluated OR event; two marginal bounds yield only their summed union bound.

The retained CART and random-forest predictors share a 1,120-source development set, split into 926 training and 194 validation sources. CART has depth seven and minimum leaf size 20; the forest averages 64 depth-eight trees with minimum leaf size 12. Causal features describe uncertainty, information, residuals, steps, iteration, pair count, displacement and declared noise. Predictors estimate terminal MSE $\widehat{v}_{ja}$, remaining work $\widehat{c}_{ja}$ and a development-event frequency $\widehat{r}_{ja}$ for reachable cap a. The endpoint heuristic selects

$$\widehat{a}_j = \arg\min_{a\in\mathcal{A}_j} \{T\max(\widehat{v}_{ja}, 0) + \lambda\max(\widehat{c}_{ja}, 0) + \omega\,\text{clip}(\widehat{r}_{ja}, 0, 1)\}, \tag{9}$$

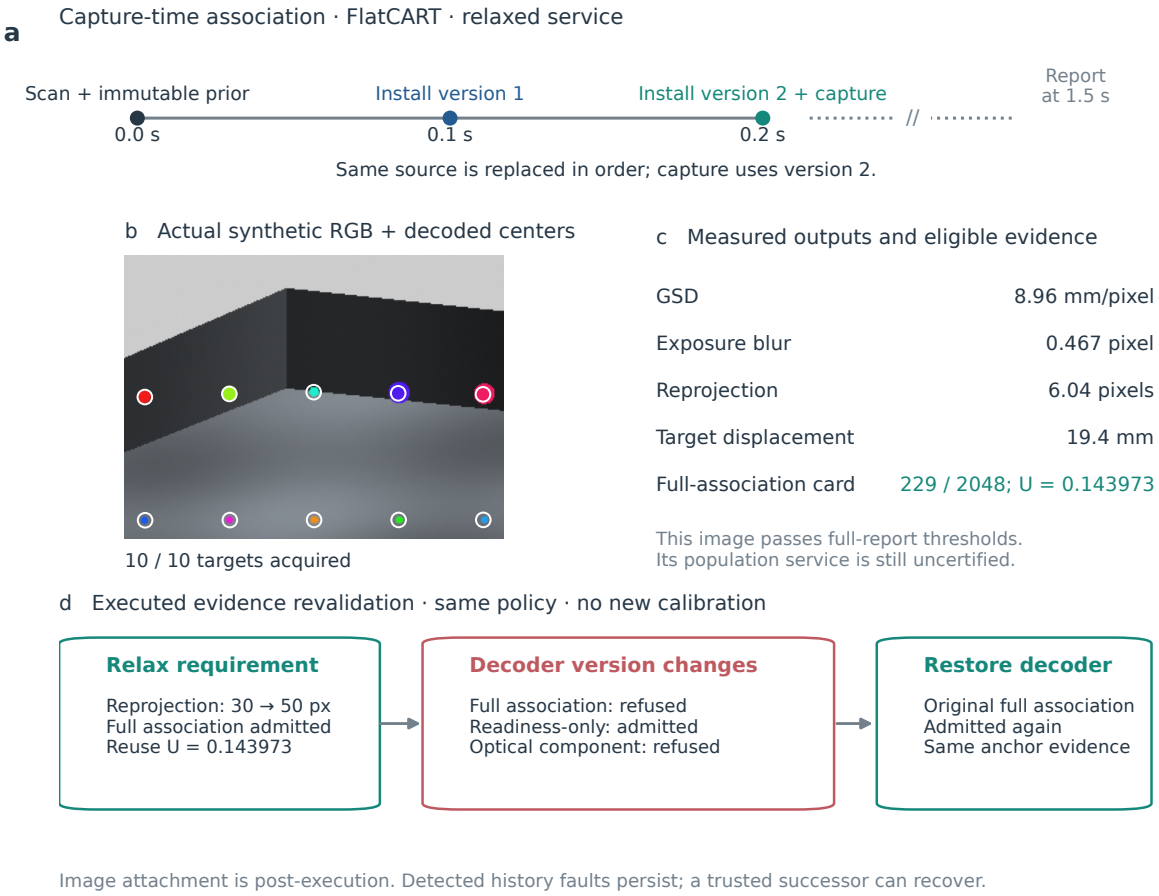


Figure 2: A complete archived source and executed software revalidation. (a) Actual numerical installations and capture use the frozen virtual clock. (b) The original synthetic RGB array with decoded-center overlays. (c) Measured outputs and the applicable full-association card. (d) Deterministic requirement/decoder fixtures reuse, withdraw and restore evidence with the policy held fixed. Panel (d) evaluates deterministic decoder-specification configurations against preloaded evidence.

with $T = 4$ s denoting the episode horizon, $\lambda \in \{0.3, 1, 3\}$ m^2 and $\omega = 0.002$ m^{2}s. MSE comparisons also guide publication of eligible candidates. These forecasts are scheduling heuristics; event calibration concerns the complete executed wrapper.

The localization catalogue contains 22 policies: matched CART/RF variants at three prices, fixed caps, final convergence, residual and randomized controls, the earlier RAVC policy and time-only interventions. We call the price-one time/profile-screened policy Contract-CART; its archived identifier is KG1. The earlier CART scheduler is retained under its archived identifier OldRAVC and specified in the supplement. FlatCART uses the same predictor; time-only and contract variants add deadline and profile predicates in sequence. Every policy receives the common next-stage budget reservation, acquisition and source-replay service. Hard-budget interruption is therefore part of each effective policy, including Fixed2.

The stage budget B covers source sealing, registration, features, forecasting, rules and publication. Total modeled work W additionally includes the common front end, receiver and stream. The frozen operation clock assigns CPU-equivalent costs to actual counted operations, then applies the declared load factor to delivery scheduling. Native CPU is a separate executed-work measurement. Selection uses the dimensionally consistent calibration objective $\overline{L} + \lambda_{\mathrm{sel}} \overline{W}$, with $\lambda_{\mathrm{sel}} = 1$ m^2. Supplementary methods give the installation-clock equation and the boundaries of component and whole-route CPU.

Table 3: Reference contract procedure. Statistical commitment precedes sensing; ordered numerical execution and evidence attachment retain distinct state.

Step	Operation
1	Compile the task into required events, engineering units and dependency projections.
2	Match the population, service and effective policy against the applicable card inventory.
3	Validate each card and its containment witness; retain candidates whose required bounds satisfy $U \le \delta$.
4	Minimize $\overline{L} + \lambda_{\mathrm{sel}}\overline{W}$ and seal the selected policy and evidence before opening the source. Return `REFUSED` for an empty catalogue.
5	Execute the retained solve under the stage reservation; publish eligible versions in source order.
6	Replace the active source contribution, replay acquired motion, and check source/version identity, dependencies and persistent history.
7	Expose the task output and its applicable evidence status; withdraw affected statements after a detected invalid history or dependency change.

3.4. Calibration, admission and evidence lifetime

A typed contract records the source, numerical version, policy, population, service, requested event and their dependencies. Before opening the scan, a request fixes its event set and risk allowance δ. For a frozen event with k failures among n iid complete calibration inputs, the stored one-sided bound is

$$U(k,n) = \begin{cases} 1, & k = n, \\ \mathrm{Beta}^{-1}(1-\beta/K; k+1, n-k), & k < n. \end{cases} \tag{10}$$

With $\beta = 0.05$ and localization family $K = 22\times2\times4\times3 = 528$, a union bound gives simultaneous risk coverage of at least $1-\beta$. Dependence between policies or events on the same source is allowed. Calibration-dependent selection can use the resulting bounds because they hold for the entire frozen family on one confidence event.

An admitted policy matches the request's population, service and dependencies and satisfies $U \le \delta$ for every required event. The selector minimizes the same calibration loss–work objective and commits before sensing. An empty admissible catalogue refuses the statistical service. During execution, receipts check source sealing, ordered version identity, active dependencies and scope history. If $A_{s,j}$ is the current attachment candidate and $M_{s,j}$ the persistent history state,

$$M_{s,j+1} = M_{s,j} \wedge \mathrm{validAttempt}_{s,j}, \qquad A^{\mathrm{issued}}_{s,j} = A_{s,j} \wedge M_{s,j+1}. \tag{11}$$

Reconstructing a plan retains this source-scoped latch. Changed dependencies revoke affected cards; unrelated cards remain usable. The attachment observer modifies claim state while preserving canonical numerical decisions, installation ordering and modeled work. Its separately measured CPU and the noninterference proof are provided in the supplement.

The bound concerns unconditional events under the frozen program and stated deployment mixture. Applying it requires matched input distributions and trusted dependencies. Source checks establish execution identity; subgroup, receipt-conditioned and route risks require additional calibration. These conditions are used consistently throughout the evaluation.

4. Changing inspection requirements

4.1. Acquisition, association and report outputs

The inspection extension retains the localization state, solver and replay interface. Its source-bound receipt observer checks frozen plans against archived execution records after numerical processing; its timing is reported separately. Up to 15 CAD face/barycentric addresses are planned before stochastic sensing. Raycasting creates RGB images with procedural texture, fixed lighting, 25-mm-radius diagnostic landmarks, a motion/exposure blur surrogate and Gaussian intensity noise. A nearest-palette decoder estimates landmark

centers from pixels and known colors. Capture-time association uses the source version actually installed at the independently sampled capture tick $c \in \{0.1, 0.2, 0.3\}$ s.

At a planned target with range d, pixel focal length f_{px} and absolute incidence γ, geometric GSD is $g = d/(f_{\mathrm{px}}\gamma)$. We score the maximum GSD and blur over fixed planned addresses, acquired coverage from decoded landmarks, reprojection error in pixels, CAD-target displacement and positive-version readiness. At speed v and exposure τ, the target blur proxy is $\max_i v\tau/g_i$; the raster uses a Gaussian exposure surrogate characterized by its pixel-median GSD. For nominal CAD target q_i and its physical counterpart $q_i^\star$, the metre-valued association quantity is

$$a_{\mathrm{assoc}} = \max_{i \in \mathcal{I}_c} \left\| \widehat{m}_R q_i + \widehat{m}_t - (m_R^\star q_i^\star + m_t^\star) \right\|, \tag{12}$$

where $\mathcal{I}_c$ contains visible in-ROI targets with finite decoded centers and positive predicted depth. Reprojection separately compares pixels predicted from the active joint state with decoded centers. These scored quantities retain distinct units and dependencies. Full acquisition requires GSD at most 20 mm/pixel, blur at most 0.5 pixel and coverage at least 0.50. Full association requires coverage at least 0.50, reprojection at most 30 pixels, displacement at most 0.10 m and readiness at capture.

Full reporting adds delay at most 2 s and image age at most 1.8 s. With the frozen virtual raster/decoder work w_{camera} and load α,

$$t_{\mathrm{report}} = \max\{t_{\text{first positive}}, c + 0.05 + \alpha w_{\mathrm{camera}}\}, \qquad a_{\mathrm{image}} = t_{\mathrm{report}} - c. \tag{13}$$

This clock characterizes the virtual producer. The declared thresholds define synthetic diagnostic services. Their complete eight-variant specification and all unit checks are supplied in the supplement.

4.2. Requirement containment and producer dependencies

A requirement R compiles into admissible intervals and a failure event:

$$S_R = \{z : \ell_{R,j} \le z_j \le u_{R,j},\ j \in J_R\}, \qquad E_{p,R} = \{z_p(X) \notin S_R\}. \tag{14}$$

Nonfinite required outputs fail; the compiler rejects invalid specifications and incompatible units. An anchor A covers R when $J_R \subseteq J_A$ and $\ell_{A,j} \ge \ell_{R,j}$, $u_{A,j} \le u_{R,j}$ for each requested quantity. The resulting witness establishes

$$S_A \subseteq S_R \quad \Longrightarrow \quad E_{p,R} \subseteq E_{p,A} \quad \Longrightarrow \quad \Pr(E_{p,R}) \le U_{p,A}. \tag{15}$$

On the simultaneous anchor-coverage event, this implication holds for the entire containment closure. Calibration-dependent anchor selection within that closure adds no separate event family. Dropping a required quantity defines a partial service and is recorded explicitly.

Producer changes require an additional condition. Let D_j be the frozen implemented dependency set of quantity j, and $D_R = \bigcup_{j \in J_R} D_j$. An event capsule identifies the component snapshot projected onto D_R. Transfer requires the requested functions to remain identical on supported complete inputs:

$$z_{p',j}(X) = z_{p,j}(X) \quad (j \in J_R) \quad \Longrightarrow \quad E_{p',R} = E_{p,R}. \tag{16}$$

Together with containment and an unchanged population, this equality preserves the old anchor bound. Producer compatibility is evaluated against the trusted implemented dependency specification. The three rules serve distinct implementation decisions: containment authorizes a changed requirement, functional invariance retains evidence after an unrelated provider change, and source-history validity permits attachment. The following theorem composes these rules into the correctness property of the operating mechanism.

Theorem 1 (Coverage of an authorized event)**.** *For one frozen calibration family, consider a request committed before sensing. Let a valid anchor A cover its requested failure event E_R, with $U_A \le \delta$. Assume the deployment population is matched and the requested producer functions remain invariant. On the simultaneous calibration-confidence event, $\Pr(E_R) \le \delta$. For an issuance indicator I generated by the source-history monitor,*

$$\Pr(E_R \cap \{I = 1\}) \le \delta. \tag{17}$$

A detected invalid attempt makes issuance false for subsequent attachments in that source scope while preserving valid numerical installation.

Proof. Requested-function invariance preserves the request event, and the containment witness gives $E_R \subseteq E_A$. Simultaneous anchor coverage yields $\Pr(E_R) \leq \Pr(E_A) \leq U_A \leq \delta$. Intersecting with the issuance event preserves the upper bound. The conjunction in Eq. (11) remains false after a failed attempt, independently of later numerical installations. □

The simultaneous confidence event has calibration probability at least $1 - \beta$. Equation (17) bounds the joint event of task failure and evidence issuance. The theorem justifies the implemented reuse and issuance decisions through the existing finite-family bound; its state property ensures that plan reconstruction preserves a detected invalid history.

A wider ROI changes acquired coverage, association outputs and camera work. At the same planned addresses, fixed pixel focal length and target geometry preserve geometric GSD. Optical evidence can therefore remain applicable while a complete image or report event requires renewal. Capture-time readiness similarly uses the version available at capture; future pose updates do not retroactively alter it. Policy and anchor commitment before sensing avoids per-image switching among marginally calibrated events.

4.3. Reuse with direct-renewal fallback

For a committed numerical policy π, let $\mathcal{A}_\pi(R)$ contain compatible anchors that cover R and satisfy its allowance, and let $\mathcal{D}_\pi(R)$ contain compatible exact-event cards. The evidence route is

$$Q_\pi(R) = \begin{cases} \text{reuse}(A), & \mathcal{A}_\pi(R) \neq \varnothing, \\ \text{direct}(D), & \mathcal{A}_\pi(R) = \varnothing,\ \mathcal{D}_\pi(R) \neq \varnothing, \\ \text{refuse}, & \text{otherwise.} \end{cases} \tag{18}$$

Both branches use the same simultaneous calibration event and are chosen before sensing. Reuse therefore adds a proof-construction option while the exact-event fallback retains directly supportable requests. The routing analysis fixes the original direct-renewal policy, separates support from proof-preparation work, and deduplicates proof identities by event, producer, population and calibration source manifest.

4.4. Fair evidence-reuse alternatives

Contract coverage uses explicit containment witnesses and dependency capsules. Complete procedural coverage independently implements the same interval and dependency semantics. Cached direct renewal evaluates predeclared requested events using applicable calibration records, including invariant pools available to the other methods. Exact-card matching is the limited anchor-identity control. All receive the same quantities, cards, confidence allocation, records, objective and request cache. We evaluate cold preparation separately from first-query validation and warm lookup. Their archived method identifiers and complete specifications are supplied in the supplement.

5. Experimental design

5.1. Independent inputs and prospective freezes

The localization study contains 8,192 calibration inputs, 4,096 independent confirmation inputs and 1,536 unsupported stress inputs. Nominal geometry mixes B767 and C172; the separate unfitted airframe is excluded from predictor training and receives matched fresh calibration. Each profile has 4,096 calibration and 2,048 confirmation sources. Forty-eight pilot sources are excluded. Numerical policies, population definitions, events and the operation clock were frozen before formal generation; cards and requests were sealed before confirmation. Before confirmation, the attachment monitor was changed to retain a failed-attempt history across plan reconstruction. Its numerical decisions, operation clock and event labels were unchanged. Versioned source and dependency records document the implementation chronology in the supplement.

The two inspection studies use independent sources and retain separate evidence. Study A renders 192×144 pixels and B renders 288×216, both with focal length 1,800 pixels and exposure 6 ms. Each

Table 4: Shared processing services and distinct virtual camera work. Stage budgets concern numerical sender work; total work includes additional common components.

Service	Load α (factor)	Deadline D (s)	Stage budget B (ms)
Relaxed	10	0.5	20
Standard	20	0.3	12
Tight	40	0.2	6
Contended	60	0.3	12
Virtual raster + decoder: A 62.5 ms; B 125 ms			

contains 4,096 calibration, 2,048 confirmation and 256 stress sources, totaling 12,800 across A and B. Invariant optical events can pool the independently identified calibration blocks; other events use the matched study. The reserved family is $K = 4,608$ with $\beta = 0.05$, of which 3,072 cards are stored. Unused reserve remains allocated. All 6,144 plans are sealed before confirmation.

Policy, service, event and request repetitions are paired executions of these sources. They add no independent sample size. Nominal mixtures sample prior deviations of 0.015/0.03/0.06 m, noise multipliers 1/2 and shape levels 0/1/2. Separate unsupported stress introduces occlusion, range degeneracy, larger prior error or map warp. The supplement records seeds, perturbations, freezes, amendments and full execution lineage.

5.2. Comparisons and measurement boundaries

Matched module interventions separate predictor decisions, time/profile predicates and attachment checks. Full localization selection is compared with a task-aware calibrated procedure using identical source/version checks. The primary 16 paired intervals cover two profiles, four contrasts and two metrics at standard service, using 32,000 source-bootstrap resamples. Pointwise and Bonferroni-adjusted percentile intervals are approximate sampling intervals; exact binomial bounds govern admission. Sensitivity, subgroups, Pareto means and stress remain descriptive.

The route study executes 36 distinct 600-s inputs: three geometries, three conditions and four realizations, with 12 methods and 299 scan opportunities per route. It measures the complete live online loop, with source generation, archive hydration and truth scoring outside the timer. Inspection component measurements and preloaded-record preparation have different boundaries. Windows thread-CPU measurements show 15.625-ms quantization, a property of the measurement clock. The normalized operation clock specifies a controlled processing service. Component CPU and complete-route CPU use separate measured boundaries. A diagnostic point-count probe evaluates the retained solver at caps 96, 192 and 384 on the five replay inputs, with three timing trials per cap and the actual available point count recorded.

Three complementary checks evaluate the evidence interface. Paired producer re-execution uses the five previously sealed replay sources, retaining CAD addresses, stochastic input, capture state and policy while rerunning geometry, rasterization, decoding and task scoring. Four declared interventions change ROI, focal length, exposure or the decoder. Two additional fault fixtures introduce ROI-dependent focal length and decoder-dependent readiness without declaring those couplings. Measured quantity differences, computed independently of the dependency rules, are compared with the declared affected set. Baseline pixels and metrics are checked against their archives. A separate deterministic matrix checks 15,360 implementation decisions against an independently coded oracle over the same declaration; it verifies protocol conformance. Finally, archived-card routing holds the direct-renewal policy fixed and evaluates reuse with direct fallback on all 320 unique contexts. These checks preserve the original population inputs and primary calibrated cards.

Figure 3 preserves the four original platform photographs as engineering context. Synthetic raster landmarks supply the evaluated image task. Complete current-study tables, original simulation views, routes, tail summaries and fault fixtures are retained in the supplement and reproducibility assets.

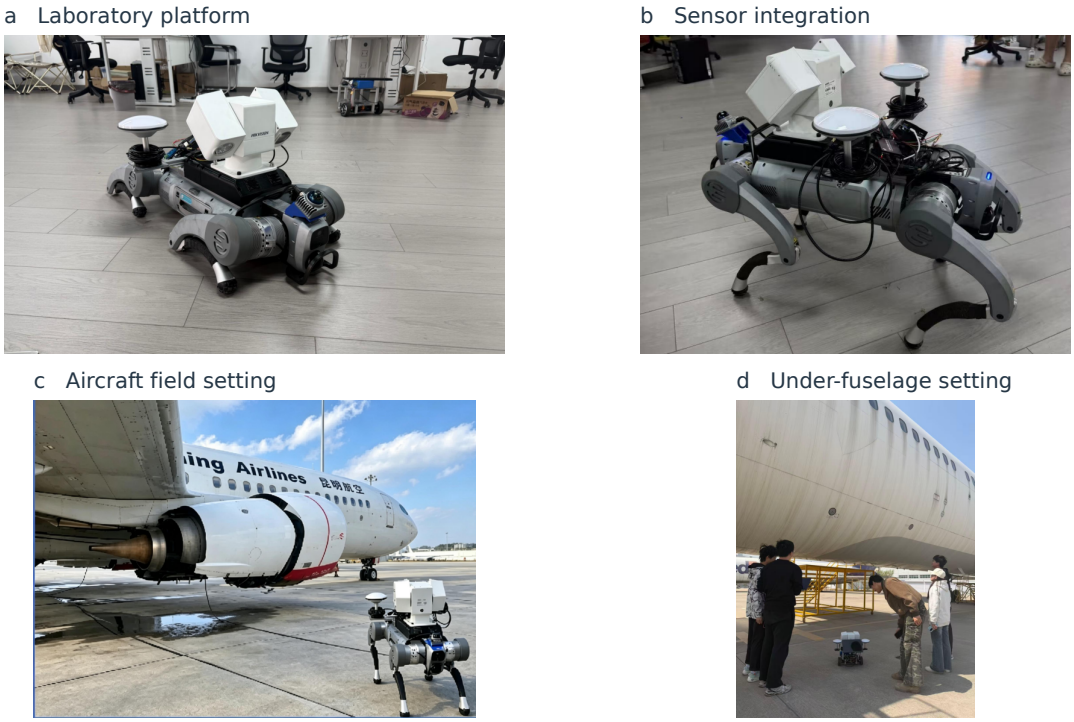


Figure 3: Original platform photographs: laboratory integration, sensor integration, aircraft field setting and under-fuselage setting. These photographs document engineering context; quantitative results in this study use the declared synthetic inputs. Original image proportions are retained.

Table 5: Primary-allowance service support. Configured labels retain the frozen request catalogue; unique contexts deduplicate compiled event identities within population and service. Full tasks retain all original requirements.

Study	Method	Labels /192	Unique /160	Full /24
A	Contract coverage	24	24	0
A	Complete procedure	24	24	0
A	Cached direct renewal	30	30	0
A	Exact-card control	18	18	0
B	Contract coverage	26	23	1
B	Complete procedure	26	23	1
B	Cached direct renewal	32	29	1
B	Exact-card control	20	17	1

6. Results

6.1. Service support under reuse and direct renewal

At allowance 0.15, anchor reuse supports 24/192 configured labels in A and 26/192 in B; direct renewal supports 30/192 and 32/192. Several variants compile to the same event. Counting unique (population, service, event) contexts gives 24/160 and 23/160 for reuse, versus 30/160 and 29/160 for renewal (Table 5). The label and unique-event counts therefore describe complementary aspects of service support. Figure 4 retains every label and the additional direct-renewal services.

The full-association primary plan for B/unfitted/relaxed uses FlatCART. It has calibration count $229/2,048$, upper bound 0.143973 and independent confirmation count $103/1,024$. The admission margin is

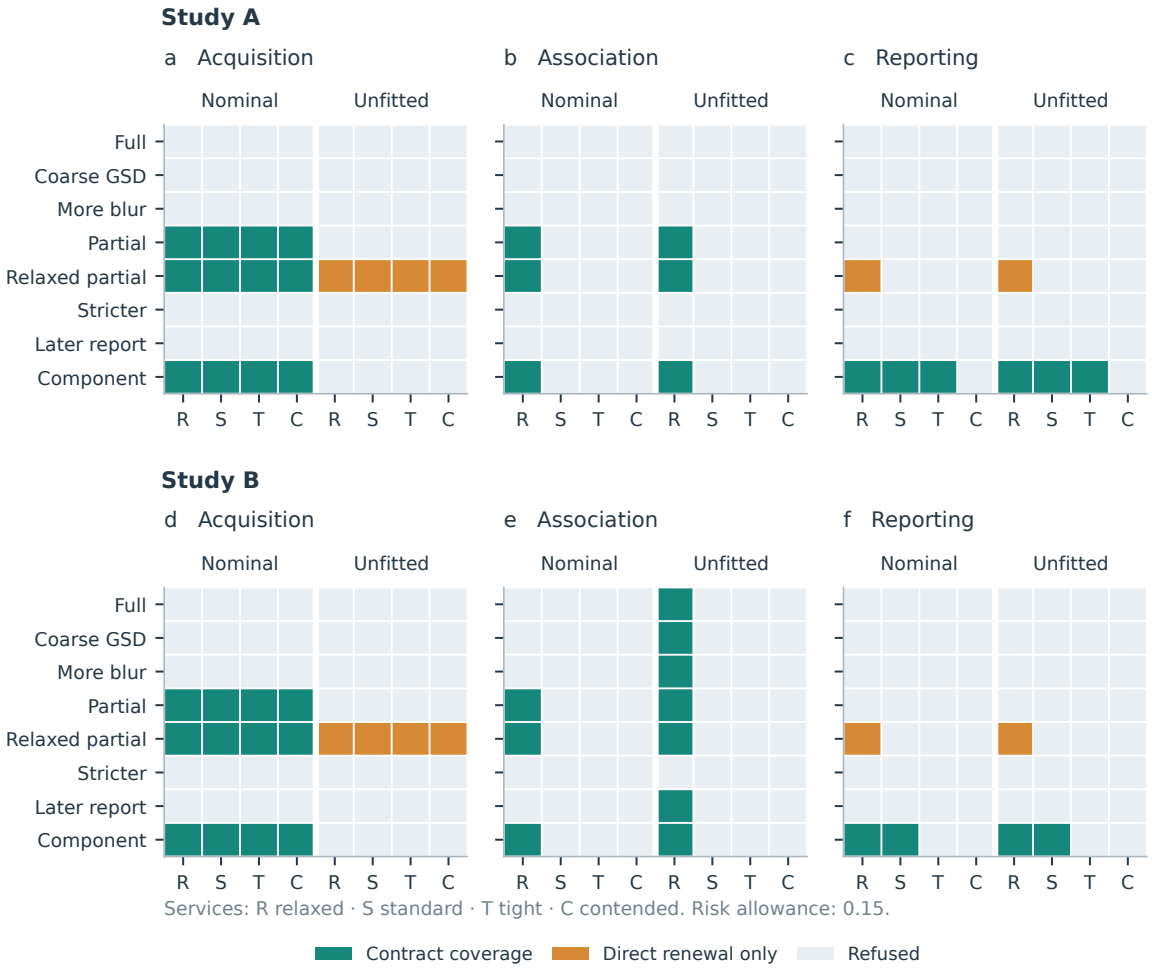


Figure 4: All 192 primary request labels per study. Teal cells show contract coverage; orange cells show additional cached direct-renewal support; grey cells are refused. Repeated event identities remain in this configured-label display and are separately deduplicated in Table 5.

0.006027. Full acquisition and reporting remain unsupported; other accepted requests are explicitly partial or component services.

6.2. Reuse-first routing and evidence-preparation work

The reuse-first route retains all 30/160 and 29/160 directly supported contexts in the two studies. With the numerical policies fixed, 46 of these 59 contexts use compatible anchors and 13 use direct-event fallback. Its selected proof catalogue contains 30 unique cards and requires 63,488 event–record evaluations, compared with 35 cards and 73,728 evaluations for exact-event preparation: a 13.89% reduction without changing supported contexts (Fig. 5a,b). This is counted preparation work for selected proofs; the inventory is preloaded and the analysis measures no new population outcome.

The original anchor-only comparison retains a different trade-off: 47 supported contexts versus 59 for direct renewal. On the 47 matched contexts and their fixed policies, it requires 49,152 rather than 61,440 record evaluations, a 20% reduction. Across the complete catalogues, preparation evaluates 2,490,368 rather than 5,373,952 records, 53.66% fewer, with narrower support. The measured cold-time difference in Table 6

Table 6: Preloaded-record timings from the original study. Cold time is the median of three trials; the first batch contains 384 queries. Warm lookup is reported in microseconds (μs), approximately 90–110 ns per cached dictionary lookup.

Method	Cards	Cold (s)	First batch (s)	Warm (μs)
Contract	1,152	0.179	6.408	0.091
Complete procedure	1,152	0.176	6.225	0.111
Cached direct renewal	2,560	0.365	9.991	0.098
Exact-card control	1,152	0.186	8.747	0.103

is 0.186 s. These small absolute times motivate retaining direct fallback; recurring warm lookups rescore no records.

Multiplicity concerns calibrated risk functions; requirements in an anchor's containment closure introduce no additional tests. The primary comparison retains the originally reserved $K = 4,608$ for every method. An archived-count sensitivity groups tests using their event, producer, population and source-manifest signatures: 780 canonical anchor tests and 1,692 tests in the complete stored family. These are alternative calibration-scope designs, each allocating its own total $\beta = 0.05$; a combined routed catalogue requires one allocation over its tested union. For the full-association example, the upper bound changes from 0.143973 to 0.140651 for the anchor design and 0.142133 for the complete design. At allowance 0.15, support remains 24/23 for anchors and 30/29 for exact events. Tighter multiplicity improves margins; it does not supply events outside the available closure. The sensitivity leaves all primary cards and plans unchanged.

6.3. Concrete source, dependency change and service recovery

An archived wide-ROI, unfitted-airframe source supplies a complete numerical and image case (Fig. 2). FlatCART installs versions one and two at 0.1 and 0.2 s; capture at 0.2 s consumes version two. All ten planned targets are acquired, reprojection is 6.04 pixels and CAD-target displacement is 19.4 mm. A prespecified consumer relaxation from 30 to 50 pixels remains covered by the original full-association anchor, reusing $U = 0.143973$ without recalibration.

Changing the decoder specification withdraws full association. A readiness-only request retains an applicable component bound $U = 0.056804$, whereas the optical component remains refused under its archived risk bound. Restoring the original decoder restores full association. This source individually passes the full-report thresholds, while the population-level full-report service remains uncertified: a favorable image does not authorize a missing population statement.

6.4. Producer re-execution and declaration auditing

Figure 6 compares actual recomputed outputs after six interventions on the five archived inputs. All baseline RGB arrays reproduce their archives, and task quantities agree within 10^{-10}. ROI changes preserve fixed-target GSD; focal-length and exposure edits alter their corresponding optical quantities. Disabling the decoder changes acquisition and association while leaving capture-time readiness available from the numerical source.

The two undeclared-coupling fixtures expose a different result. ROI-dependent focal length creates nine undeclared quantity changes across five source pairs; decoder-dependent readiness creates five. The numerical audit flags every source pair in both fixture classes, using output differences rather than the declaration oracle. Its update gate vetoes the tested candidate when an undeclared quantity change is observed, pending a corrected specification and applicable evidence. These finite re-executions test producer compatibility; the functional identity in Eq. (16) remains the transfer condition for the supported program and input domain.

Protocol conformance is evaluated separately. In the declared-model matrix, 705 edits of initially admitted contexts produce 489 withdrawals and 216 retained statements; all restorations and oracle checks agree. The complete 15,360-decision matrix, invalid-packet checks and source-history fixtures are provided in the supplement as implementation verification.

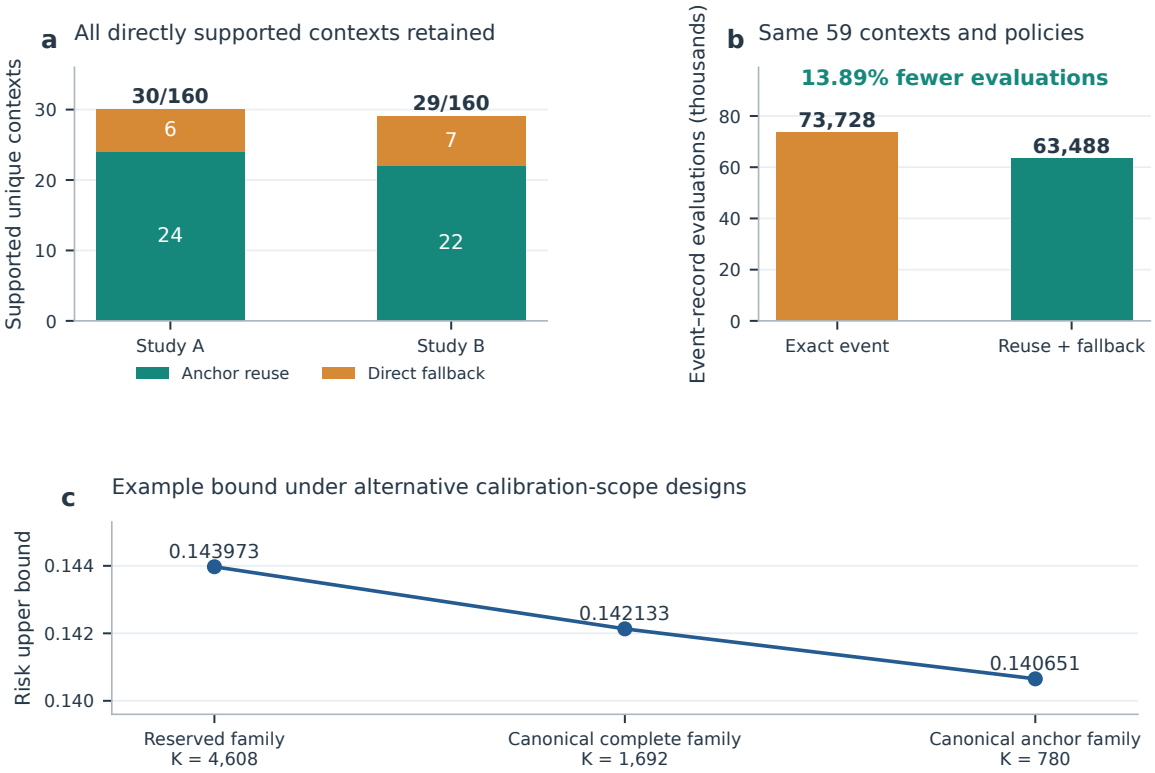


Figure 5: Evidence routing, proof preparation and calibration scope. (a) Reuse-first routing retains all 59 directly supported contexts: 46 reuse anchors and 13 use exact-event fallback. (b) With the same contexts and policies, selected proof preparation requires 63,488 versus 73,728 event–record evaluations, 13.89% fewer. (c) Alternative scope designs tighten the example risk bound while the allowance-0.15 support counts remain unchanged. Primary plans retain their reserved family.

6.5. ROI changes separate geometric quality from delivery feasibility

The excluded paired pilot in Fig. 7 uses the same source and landmarks for both ROIs. Across 48 paired pilots, fixed-target GSD is invariant: wider ROI changes acquired coverage and virtual production work while range, incidence and focal length remain fixed. The formal A/B confirmation inputs are independent; their empirical differences are not a paired causal estimate. Optical projection invariance permits selective reuse, while affected coverage, association and report events require their own evidence.

The reporting clock also creates an analytical infeasible region. Equation (13) gives

$$a_{\text{image}} \geq 0.05 + \alpha w_{\text{camera}}. \tag{19}$$

This lower bound exceeds the full-task 1.8-s limit for A tight/contended and B standard/tight/contended: five of eight study/service combinations. Their reporting failure is built into the declared timing model. The other three combinations permit timing but can fail acquisition, association or actual delivery conditions. All full-report requests remain refused at the primary allowance.

6.6. Localization accuracy, work and availability

At standard service, Contract-CART reduces integrated loss relative to Fixed2 by 15.68% on nominal geometry and 12.61% on unfitted geometry, with work increases of 16.09% and 16.07% (Table 7). Fixed4 and

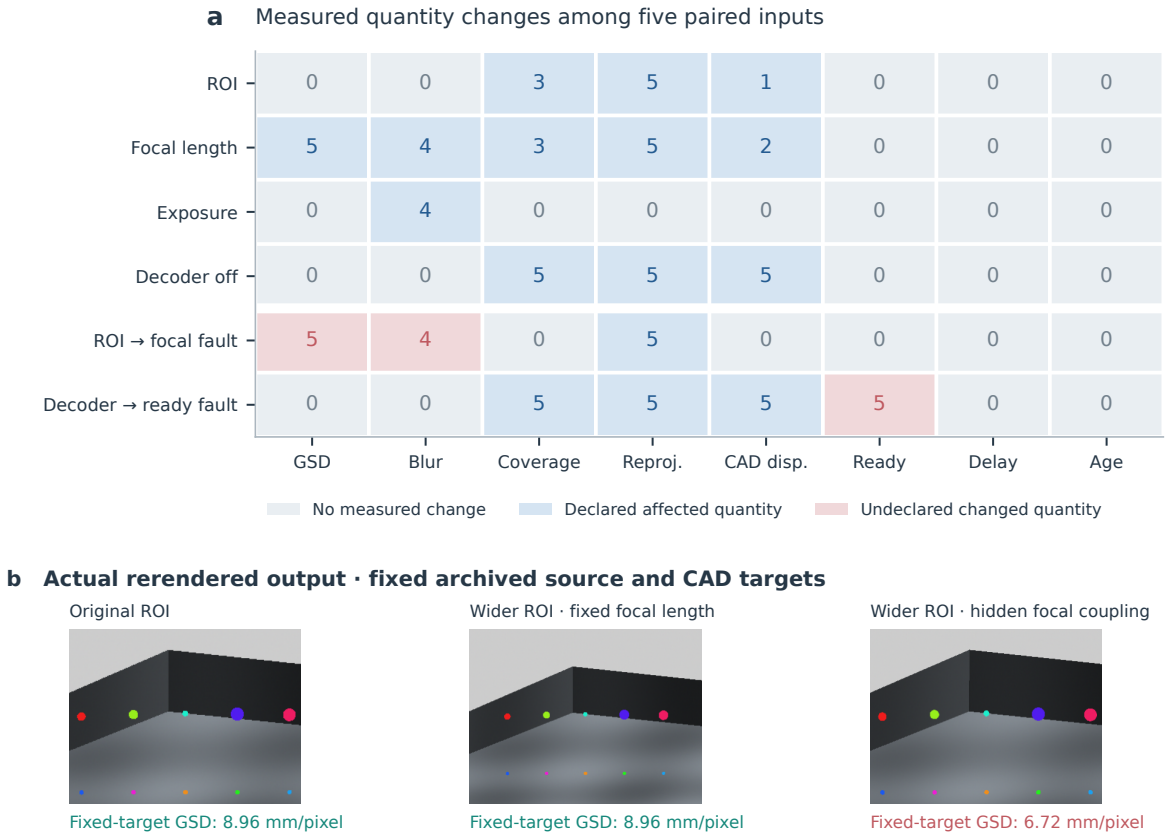


Figure 6: Actual producer re-execution on five archived inputs. (a) Each cell reports how many paired inputs change a task quantity; blue changes are covered by the declared dependency set, red changes expose an omitted coupling, and grey cells show no measured change. (b) Rerendered examples separate correct ROI expansion from an ROI-dependent focal-length fault. The checker compares numerical outputs independently of the declared dependency table.

Final further reduce mean loss at greater work, while the RF policy remains close to Fixed2. Figure 8 displays all 22-policy means and the prespecified paired intervals, covering the four matched method contrasts.

At the primary allowance, Full admits 18 of 32 localization request contexts. Relaxed and standard services admit all four request types. Contention admits trajectory-only requests; tight service admits none. Refused requests remain in support denominators and retain the propagated prior during the corresponding execution.

The gains also depend on episode difficulty. The point-count probe runs the same source solver at caps 96, 192 and 384, clipping to the available frontend cloud. Across the five inputs, actual counts range from 96 to 384 and native CPU medians span approximately 6–18 ms. Source-pose errors are nonmonotonic in point count; the supplement reports each execution and acceptance decision. These probes characterize the reference solver and retain the primary 96-point policies.

The two largest nominal contributions account for 27.99% of the Contract-CART–Fixed2 net loss gain; the unfitted share is 10.02%. Diagnostic removal leaves positive mean gains of 0.001563 and 0.002265 m^2s, while the primary comparison retains every source. Complete tail and unsupported-stress results are provided alongside these means.

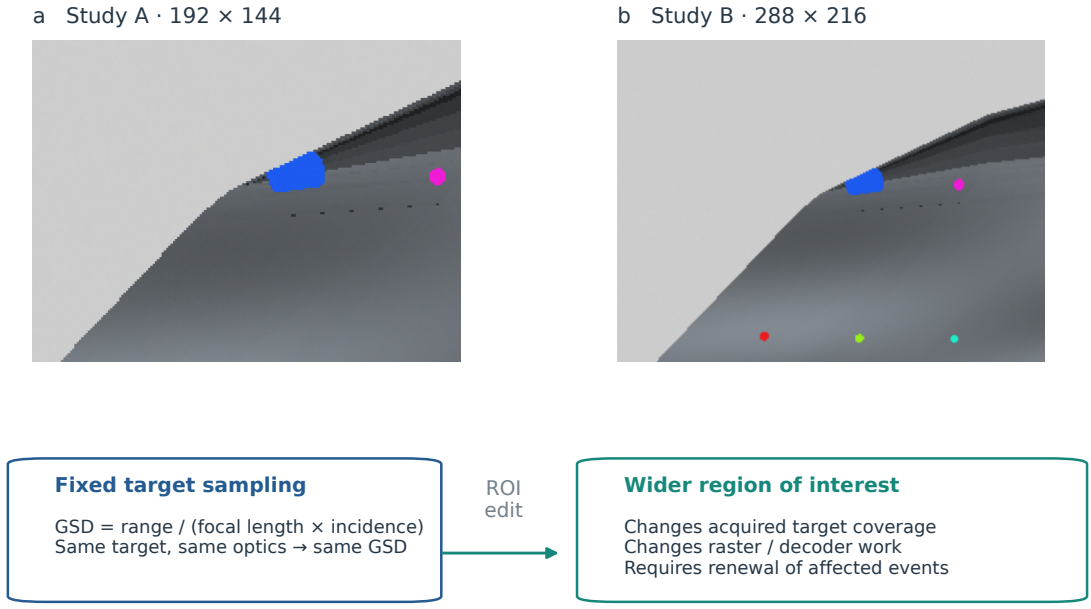


Figure 7: Actual archived synthetic raster output from excluded pilot 12000000 under the paired ROI intervention, with diagnostic landmarks and source fixed. The original displayed rasters are reproduced from their archived figure assets. Wider ROI changes coverage and work; fixed-target GSD follows the unchanged optical/geometry projection.

Table 7: Standard-service confirmation, $n = 2,048$ per population. L is integrated loss in 10^{-3} m^{2}s; W is total modeled work.

	Nominal			Unfitted		
Policy	L	W (ms)	Union fail (%)	L	W (ms)	Union fail (%)
Fixed2	13.833	9.221	8.64	19.938	9.220	10.06
Fixed4	11.295	12.109	7.52	17.350	11.736	9.28
Final	11.295	12.109	7.52	17.350	11.736	9.28
FlatCART	11.664	10.705	6.84	17.423	10.702	8.79
Contract-CART	11.664	10.705	6.84	17.423	10.702	8.79
FlatRF	13.844	9.973	8.64	19.936	9.953	10.06
ContractRF	13.844	9.973	8.64	19.936	9.953	10.06
Earlier CART	12.199	9.666	6.79	17.964	9.781	9.03
Full	12.301	9.721	7.13	17.964	9.781	9.03
Complete procedure	12.301	9.721	7.13	17.964	9.781	9.03

6.7. Action predicates and evidence checks

Time screening changes every CART stopping endpoint under contention. On nominal confirmation it reduces work by 3.600 ms and observation failure by 1.660 percentage points while increasing loss by 0.005094 m^{2}s. The unfitted changes are -3.583 ms, -1.904 percentage points and $+0.004179$ m^{2}s. The common next-stage reservation limits refinement under the tight budget. The supplement plots all-service time-predicate effects and the original attachment/dependency fixtures.

Evidence checks affect attachment correctness. Across 16 constructed invalid cases using actual frozen cards, stripped checks issue 16 incorrect attachments. Matching reduces this to four, current-version validation to two, and persistent source history to zero. The valid controls retain their attachments. Dependency edits

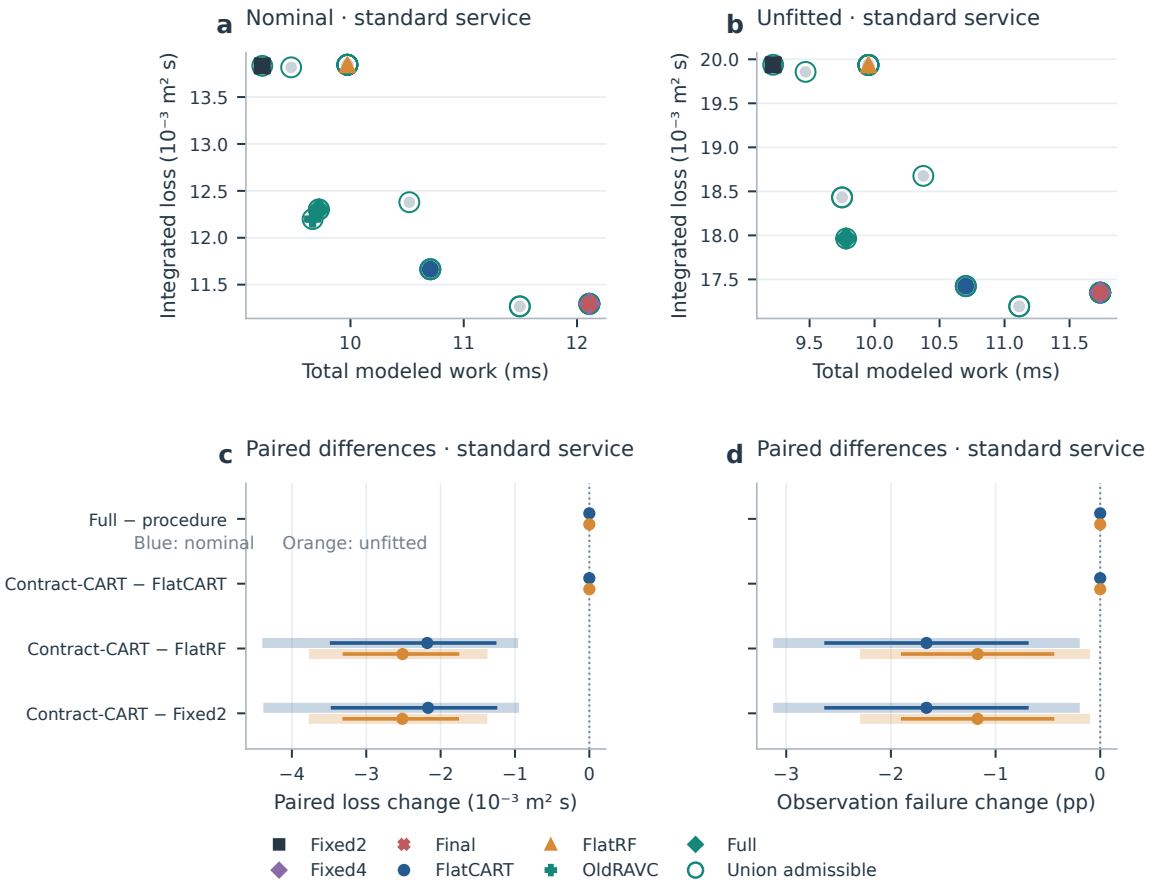


Figure 8: Standard-service localization. (a,b) All 22 policy means, highlighted controls and full selection; green rings mark direct-union eligibility at allowance 0.15. (c,d) All 16 paired intervals, colored by population: thin intervals are pointwise 95%, pale intervals are family-adjusted percentile intervals.

selectively revoke cards: changing the RF forecast affects 168 of 528, whereas a shared clock change affects all 528. These fixed software fixtures measure compliance with the specified protocol.

6.8. Continuous-route accuracy and availability

Figure 9 shows all 12-method route summaries and highlights Fixed2, Contract-CART and Full. Ordinary routes often favor Fixed2 on CPU or accuracy; modest refinement opportunities provide limited return on forecasting and repeated publication. Under C172 contention, Contract-CART raises timely-correction success from 53.01% to 80.94%, with RMSE increasing from 5.237 to 5.341 cm. Full achieves 81.61% and 5.234 cm. Whole-online CPU is 2.121, 2.194 and 2.028 s, respectively, over each 600-s route.

Severe stress produces large route errors across the catalogue. The route selector refuses episode-to-route attachments and continues an explicit provisional recommendation. All 432 executions remain descriptive route evidence; their samples and policy switches are separate from episode calibration. Full tables and the predeclared continuous trace are supplied in the supplement.

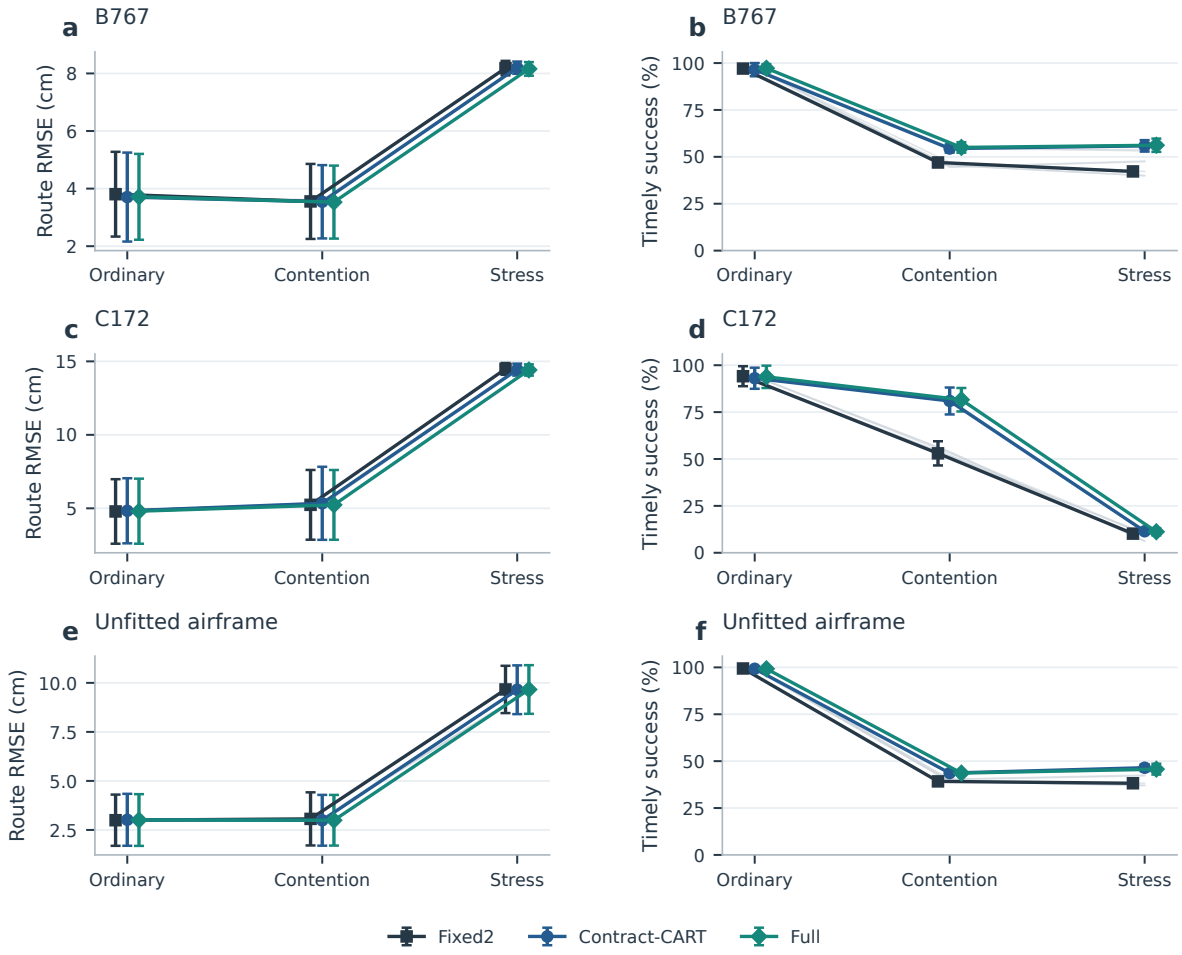


Figure 9: Complete-route results by geometry and condition. Grey traces retain all 12 methods; colored traces highlight Fixed2, Contract-CART and Full. Each point summarizes four matched 600-s inputs, with sample-standard-deviation bars. Route summaries carry no transferred episode bound.

7. Discussion

The operating contract expresses evidence applicability as an executable engineering relation. A unit-typed request compiles into an event, providers and a committed plan. A containment witness identifies an applicable statistical statement; ordered source replacement and persistent history connect it to the actual output. The contribution is the composition of these operations into an inspectable lifecycle across estimation, acquisition, association and reporting.

The three validation layers answer different questions. Numerical and image studies characterize the service outputs. Paired producer re-execution checks how changed functions affect their quantities and can expose a declaration that misses an implemented coupling. The deterministic matrix checks whether the software follows its declared rules. Keeping these layers distinct links representation, program behaviour and statistical evidence without assigning population meaning to fixture counts.

Reuse and direct renewal form complementary evidence routes. A requested event inside the anchor closure can use an existing proof; an exact-event fallback preserves requests outside that closure. On all 59 directly supported contexts and fixed numerical policies, this arrangement reduces selected proof-preparation evaluations by 13.89%. The absolute preparation times in this implementation are small; the practical

capability is to retain applicable evidence and identify precisely what must be renewed when a requirement or producer changes.

The representation guides engineering decisions through distinct quantities. ROI affects acquired coverage while fixed-target GSD follows the optical projection. The report clock identifies timing-infeasible services before risk-card lookup. Larger local registration samples alter both computation and acceptance, while the primary numerical catalogue supplies accuracy–work–availability choices. These relationships connect a requested service to the producer or evidence change needed to support it.

8. Conclusion

Executable operating contracts connect changing inspection requirements to the applicability and lifetime of statistical evidence. A typed metamodel, unit-aware compiler and executable predicates bind task events, committed policies, source versions, producer dependencies and calibrated cards. Actual paired producer re-execution complements declared-rule conformance checks. Reuse with exact-event fallback preserves all 59 directly supported contexts and reduces selected proof-preparation evaluations from 73,728 to 63,488 under fixed policies. The framework provides a traceable evidence lifecycle for aircraft localization and inspection, supported by independent numerical studies and reproducible implementation diagnostics.

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Data and code availability

The reproducibility release retains protocols, independent source seeds, native solver and predictors, execution records, risk cards, versioned protocol history, full result tables and editable source. The manuscript package includes table snapshots, paired producer re-execution inputs and code, the typed compiler, the deterministic change/recovery validator, SHA-256 provenance and reproducible PDF/SVG scripts.

Declaration of generative AI and AI-assisted technologies

OpenAI ChatGPT/Codex assisted code development, computational analysis, manuscript revision and data-figure scripts. Submitted figures use original photographs, executed synthetic raster output and deterministic vector graphics.

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Appendix A. Evidence scope and implementation context

Baseline equivalence and attribution. The standard-service KG1 numerical gains are shared by FlatCART; Fixed4/Final and FlatRF/ContractRF also have coincident reported means. FlatCART, time-only CART and KG1 have identical observed outcomes at relaxed/standard service. Profile predicates change none of the reported confirmation outputs; the first two thresholds lie outside the ordinary noise/prior ranges, while the unfitted rule and common stage reservation jointly limit refinement. Time-screen effects occur under contention. These observed coincidences do not establish a global policy identity over every supported input and service, so the original 22-policy, $K = 528$ family remains frozen.

Full and the complete procedure reproduce the numerical and receipt outcomes, including 18/32 localization admissions. Contract and complete procedural coverage agree across the 6,144 image-task plans and confirmation decisions, and share anchor accounting. Their conformance decisions also agree under the declared dependency oracle. The controls attribute numerical effects to scheduling and evidence effects to implemented interval, dependency and history semantics. The tables, paired intervals and complete catalogues preserve these results.

Statistical statements and diagnostic verification. Population-event bounds assume iid complete calibration inputs, the frozen effective program, a matched deployment mixture and trusted implemented dependencies. Theorem 1 bounds the joint event of failure and issuance; receipt-conditioned, subgroup and route risks require separate calibration. Containment transfers a valid anchor bound, while producer transfer requires functional identity on the supported input domain.

The five-source producer audit reruns concrete geometry/raster/decoder interventions. Numerical mismatches falsify a declared transfer on the tested path; agreement on five inputs does not establish global invariance or completeness for arbitrary undeclared couplings. The two injected fault classes measure the specified diagnostic. The 705-edit matrix instead verifies declared-rule implementation. Fresh trusted execution scopes reuse the same archived source seal, establishing scope-level recovery. Calibration-scope sensitivity compares alternative families from existing counts; primary cards retain their original allocation.

Physical implementation and measurement scope. The quantitative studies use declared synthetic localization, route and image inputs. Original photographs supply platform context. Physical implementation requires representative observations and platform-specific timing with matched population calibration; human maintenance effort and physical fault rates have their own engineering endpoints. The virtual operation clock, Windows component CPU, complete-route CPU and preloaded-record query times describe distinct boundaries. Point-count probes report native Linux CPU medians from three repetitions per archived input and do not certify larger-point policies under the primary budget model. Full reporting remains refused at the primary allowance; satisfying that complete requirement calls for an appropriate producer service and calibration.